

● EASY

● ALGEBRA

● ANSWER KEY

Mixed

10 answers · Rationale-rich · Teach from doc

03

Q1**B****B.** 38

Choice B is correct. It's given that function f is defined by $f(x) = 80 - 6x$. The value of $f(7)$ can be found by substituting 7 for x in the given function, which yields $f(7) = 80 - 6(7)$, or $f(7) = 80 - 42$, which is equivalent to $f(7) = 38$. Therefore, the value of $f(7)$ is 38.

Choice A is incorrect. This is the value of $80 - 6(7)$, not $80 - 67$.

Choice C is incorrect. This is the value of $80 - 6(1)$, not $80 - 67$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 67$.

Q2**D****D.** $y = 4.8x + 162$

Choice D is correct. It's given that the length of the whale was 162 cm when it was born and that its length increased an average of 4.8 cm per month for the first 12 months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in cm, is 4.8 times x , or $4.8x$. The length of the whale y , in cm, can be found by adding the whale's length at birth, 162 cm, to the total increase in length, $4.8x$ cm. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q3**4**

The correct answer is 4. A solution to a system of equations must satisfy each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy -plane of each equation in the system. According to the graph, the point x, y that lies on the graph of each equation in the system is 4, 1. Therefore, the solution to the system is 4, 1. It follows that the value of x is 4.

Q4**C**

C. 80

Choice C is correct. It's given that the function f is defined by $fx = 25x + 30$. Substituting 2 for x in this equation yields $f2 = 252 + 30$, which is equivalent to $f2 = 50 + 30$, or $f2 = 80$. Therefore, the value of fx is 80 when $x = 2$.

Choice A is incorrect. This is the value of 252, not $252 + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $252 + 30$.

Choice D is incorrect. This is the value of $25 + 302$, not $252 + 30$.

Q5**348**

The correct answer is 348. It's given that 1 fathom is equivalent to 6 feet. Therefore, 58 fathoms is equivalent to $58 \text{ fathoms} \frac{6 \text{ feet}}{1 \text{ fathom}}$, or 348 feet. Thus, when the camera is at a depth of 58 fathoms, the camera's depth, in feet, is 348.

Q6**A****A. \$551.25**

Choice A is correct. At the exchange rate of 1 peso = 0.075 US dollars, 7,500 pesos would be converted to $7,500 \times 0.075 = \$562.50$. However, since Maria pays a 2% fee on the converted US dollar amount, she receives only $(100 - 2)\%$, or 98%, of the converted US dollars, and $562.50 \times 0.98 = \$551.25$.

Choice B is incorrect. This is the number of US dollars Maria would receive if the exchange service did not charge a 2% fee. Choice C is incorrect and may result from a decimal point error made when calculating the conversion to US dollars and from not assessing the 2% fee. Choice D is incorrect and may result from reversing the units of the exchange rate.

Q7**D****D. Range of the high temperatures**

Choice D is correct. The range of a data set is the difference between the maximum and the minimum values in the set. The maximum value among the high temperatures in the table is 82°F and the minimum value is 80°F . Therefore, the range is $82^{\circ}\text{F} - 80^{\circ}\text{F} = 2^{\circ}\text{F}$.

Choice A is incorrect. The median of a data set is the middle value when the values in the set are ordered from least to greatest. Ordering the high temperatures this way gives the list 80, 81, 81, 81, 82. Therefore, the median high temperature is 81°F . Choice B is incorrect. The mean high temperature is $\frac{81 + 80 + 81 + 81 + 82}{5} = \frac{405}{5} = 81$. Choice C is incorrect. The mode is the value that occurs the greatest number of times. For the set of high temperatures shown, 81 is the value that occurs 3 times, and therefore, 81°F is the mode of the high temperatures.

Q8**B****B.** $13w$

Choice B is correct. Combining like terms inside the parentheses of the given expression, $20w - 4w + 3w$, yields $20w - 7w$. Combining like terms in this resulting expression yields $13w$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**B****B.** 30

Choice B is correct. Multiplying the left- and right-hand sides of the given equation by 25 yields $x^2 = 900$. Taking the square root of the left- and right-hand sides of this equation yields $x = 30$ or $x = -30$. Of these two solutions, only 30 is given as a choice.

Choice A is incorrect. This is a solution to the equation $x^2 = 36$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

A

A. 9

Choice A is correct. The area A , in square inches, of a rectangle is the product of its length l , in inches, and its width w , in inches; thus, $A = lw$. It's given that the area of a rectangle is 630 square inches and the length of the rectangle is 70 inches. Substituting 630 for A and 70 for l in the equation $A = lw$ yields $630 = 70w$. Dividing both sides of this equation by 70 yields $9 = w$. Therefore, the width, in inches, of this rectangle is 9.

Choice B is incorrect. This is the length, not the width, in inches, of the rectangle.

Choice C is incorrect. This is half the area, in square inches, not the width, in inches, of the rectangle.

Choice D is incorrect. This is the difference between the area, in square inches, and the length, in inches, of the rectangle, not the width, in inches, of the rectangle.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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